

SATAG: SUPERPOSITION-AWARE TEMPORAL AUDIO GENERATION

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ABSTRACT

Recent advances in text-to-audio (TTA) generation have enabled more precise control over the timing of individual sound events. However, generating events with overlapping time intervals can result in inaccurate event boundaries, missing events, or sounds that do not match the requested events. We attribute these problems to the lack of explicit modeling of acoustic superposition when multiple events overlap. To address these problems, we propose SATAG, Superposition-Aware Temporal Audio Generation. SATAG detects temporal overlap and uses a temporally controllable audio generation network to generate event-specific waveforms. It then adjusts their relative amplitudes, combines them through waveform-domain superposition, and applies peak normalization to the final mixture. Quantitative evaluation shows that SATAG reduces onset and offset errors for detected events under overlapping conditions compared with existing temporally controllable TTA models. Subjective evaluation further shows that SATAG improves Quality MOS and Temporal MOS by 23.2% and 30.5%, respectively, compared with DegDiT. A demo page is available at <https://julv0207.github.io/SATAG/>.

Index Terms— Audio Generation, multi event, Temporal control, acoustic superposition, training free strategy

1. INTRODUCTION

Text-to-Audio (TTA) generation has rapidly advanced toward generating diverse environmental sounds, sound effects, and complex acoustic scenes from natural-language descriptions. Recent studies have focused on improving temporal control in TTA by incorporating event-level temporal information into the generation process. For example, DegDiT [1] represents sound events and their temporal relations using a graph structure to control event timing, and PicoAudio2 [2] uses event-level onset and offset timestamps to control event timing. These approaches provide fine-grained temporal control for individual events.

However, we find that temporally controllable TTA models struggle to generate overlapping events with the intended acoustic characteristics and timing. As shown in Fig. 1, DegDiT follows the specified timing for the non-overlapping events in (a) and (b). Under the overlapping conditions in (c) and (d), however, an event may extend beyond its specified interval, or the generated sounds may not match the requested events. (e) and (f) show the corresponding SATAG examples with the same onset and offset timestamps as (c) and (d).

Under the linear acoustics assumption, overlapping sounds can be represented as a sum of source waveforms. Existing temporally controllable TTA models jointly generate multiple sound events in a single generation process, conditioned on their descriptions and

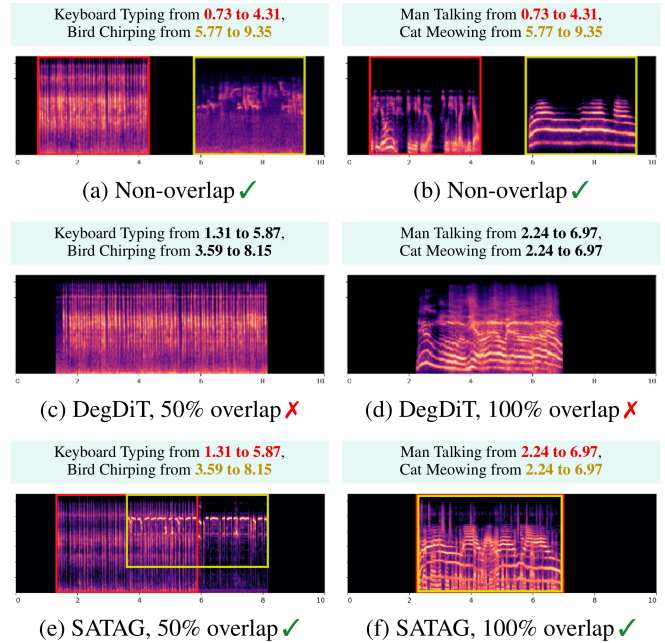


Fig. 1. Mel spectrograms of DegDiT (a–d) and SATAG (e–f). In (c), bird chirping is missing; in (d), the two sounds are incorrectly blended.

timestamps. This generation process does not explicitly enforce waveform superposition, which may contribute to missing events, inaccurate boundaries, and distorted event characteristics when multiple events occur simultaneously.

To address these problems, we propose Superposition-Aware Temporal Audio Generation (SATAG), a training-free inference framework that combines temporally controlled event generation with waveform-domain superposition. SATAG checks whether the event intervals overlap. For non-overlapping inputs, it retains the underlying model’s generation procedure. When event intervals overlap, SATAG generates each event separately under its specified timing constraints and combines the resulting waveforms through sample-wise addition. SATAG also allows relative amplitude adjustment for individual events. SATAG requires no additional training or changes to the underlying model.

Our contributions are as follows: (1) we propose SATAG, a training-free framework for generating overlapping sound events through separate event generation and waveform-domain superposition, with relative amplitude control; (2) we construct AudioCapsT by adding automatically estimated event-level onset and offset timestamps to AudioCaps; and (3) we evaluate SATAG against existing temporally controllable TTA models and demonstrate strong performance in both quantitative and subjective evaluations.

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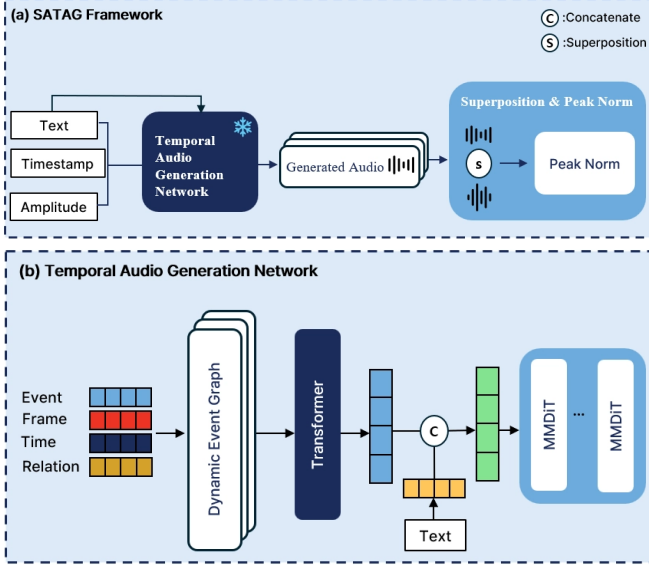


Fig. 2. Overall framework of SATAG and temporal audio generation network.

2. BACKGROUND

2.1. Principle of Superposition

The principle of superposition is a fundamental property of linear systems, stating that the response to a linear combination of multiple inputs is equal to the same linear combination of their individual responses. Let $\mathcal{L}$ denote a linear operator, and let u_1 and u_2 be the responses corresponding to inputs f_1 and f_2 , respectively:

$$\mathcal{L}[u_1] = f_1, \quad \mathcal{L}[u_2] = f_2.$$

For arbitrary constants a and b , the linearity of $\mathcal{L}$ gives

$$\mathcal{L}[au_1 + bu_2] = a\mathcal{L}[u_1] + b\mathcal{L}[u_2] = af_1 + bf_2. \quad (1)$$

Thus, the response to the combined input $af_1 + bf_2$ can be expressed as

$$u = au_1 + bu_2. \quad (2)$$

More generally, for N inputs, $f = \sum_{i=1}^N a_i f_i$, the corresponding response is

$$u = \sum_{i=1}^N a_i u_i. \quad (3)$$

Accordingly, a complex input can be decomposed into multiple components, whose individual responses are calculated independently and linearly combined to obtain the overall system response. In audio signals, the same principle provides the basis for representing simultaneous sound sources through the linear combination of their corresponding waveforms.

2.2. Temporally Controllable TTA

Temporally controllable TTA uses event descriptions and timing information to control what sounds are generated and when they occur. Existing methods represent timing conditions through dynamic event graphs in DegDiT [1], timestamp matrices in PicoAudio2 [2], and natural-language descriptions in AudioComposer [3]. Temporal control also extends to the generation procedure. ControlAudio [4] uses progressive conditioning and guidance, and FreeAudio [5] com-

bines temporal planning, attention control, and latent composition in a training-free framework.

These methods provide different ways to specify and control event timing. However, specifying the temporal placement of individual events does not explicitly impose waveform-domain superposition when their intervals overlap.

3. SUPERPOSITION-AWARE TEMPORAL AUDIO GENERATION

3.1. Temporal Audio Generation Network

We adopt DegDiT [1] as the underlying temporally controllable TTA model. DegDiT uses dynamic event graphs to encode event descriptions, timing, and inter-event relations. Following its formulation, the description c_i and onset and offset timestamps (s_i, e_i) of event i are encoded as

$$\begin{aligned} \mathbf{e}_i^{\text{type}} &= \text{Proj}(\text{MeanPool}(\text{T5Encoder}(c_i))), \\ \mathbf{t}_i &= \text{MLP}([s_i, e_i]), \\ \mathbf{f}_i &= \text{FrameEncoder}(\mathbf{F}_{i,:}). \end{aligned} \quad (4)$$

Here, $\mathbf{F}_{i,:}$ represents frame-wise temporal activity. Inter-event relations are represented by $\mathbf{A}_{i,j,:}$, which encodes before, after, overlap, contains, and contained-by relations. For a nonempty set $\mathcal{N}_i$ of other events, DegDiT computes the relation embedding as

$$\mathbf{r}_i = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} \text{RelationEncoder}(\mathbf{A}_{i,j,:}). \quad (5)$$

These embeddings are fused and processed by a transformer encoder.

$$\begin{aligned} \mathbf{g}_i &= \text{Fusion}([\mathbf{e}_i^{\text{type}} + \mathbf{t}_i, \mathbf{r}_i + \mathbf{f}_i]), \\ \mathbf{H} &= \text{TransformerEncoder}(\mathbf{G} + \mathbf{P}, \mathbf{M}), \end{aligned} \quad (6)$$

where $\mathbf{G}$ stacks the event representations, $\mathbf{P}$ contains positional embeddings, and $\mathbf{M}$ masks padded events. The graph embeddings $\mathbf{H}$ are combined with text embeddings $\mathbf{E}_{\text{text}}$ and duration embeddings $\mathbf{E}_{\text{dur}}$ to form the generation condition

$$\mathbf{E}_{\text{cond}} = \text{Concat}(\mathbf{E}_{\text{text}}, \mathbf{H}, \mathbf{E}_{\text{dur}}). \quad (7)$$

DegDiT is trained with the flow-matching objective

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{\mathbf{x}_1, \mathbf{x}_0, t} [\|\epsilon_\theta(\mathbf{z}_t, t, \mathbf{E}_{\text{cond}}) - (\mathbf{x}_0 - \mathbf{x}_1)\|_2^2], \quad (8)$$

where $\mathbf{z}_t = (1-t)\mathbf{x}_1 + t\mathbf{x}_0$, $\mathbf{x}_1$ is a clean audio latent, $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$ is noise, and $t \in [0, 1]$. The network ϵ_θ predicts the target velocity $\mathbf{x}_0 - \mathbf{x}_1$.

SATAG uses the trained DegDiT model without further parameter updates. For overlapping events, it applies the model separately to each event. We denote the complete generation process, including latent sampling and waveform decoding, by $\mathcal{G}_\theta$ and write

$$\hat{x}_i = \mathcal{G}_\theta(\mathbf{C}_i; \xi_i), \quad (9)$$

where $\mathbf{C}_i$ denotes the full conditioning input for event i , including its description and onset and offset timestamps, and ξ_i denotes the sampling noise.

3.2. Superposition and Peak Normalization

SATAG combines separately generated event waveforms through waveform-domain superposition. Since independently generated waveforms may exhibit different amplitude scales, direct superposition can result in an unintended imbalance between sound sources. We therefore normalize each waveform to a unit peak amplitude before applying the relative mixing weights, treating waveforms with peaks at or below ϵ as silent.

Given the generated waveform $\hat{x}_i$ of the i -th event, source-wise peak normalization is defined as

$$\tilde{x}_i = P(\hat{x}_i), \quad P(x) = \begin{cases} x/p(x), & p(x) > \epsilon, \\ 0, & p(x) \leq \epsilon, \end{cases} \quad (10)$$

where $p(x) = \max_n |x[n]|$, n denotes the sample index, and $\epsilon = 10^{-8}$ prevents division by zero.

We then apply a relative mixing weight $r_i > 0$ to each normalized waveform and sum the weighted waveforms:

$$x_{\text{mix}}[n] = \sum_{i=1}^N r_i \tilde{x}_i[n]. \quad (11)$$

Finally, we apply peak normalization as $\hat{x}_{\text{SATAG}} = P(x_{\text{mix}})$ to keep the output waveform within $[-1, 1]$. For mixtures with a peak above ϵ , this common scaling preserves the relative mixing weights.

4. EXPERIMENTS

4.1. Dataset

To train DegDiT for event-level temporal control, we use audio-text pairs with event descriptions and onset and offset timestamps. AudioSet-Strong [6, 7] pairs event timestamps with predefined class labels. AudioCondition [8] combines these temporal annotations with captions from WavCaps [9], but its temporal annotations remain tied to the predefined classes. A class label alone does not capture all the acoustic details of an event. For example, the label *Dog* does not distinguish dog whinnying from dog barking.

To address this limitation, we construct AudioCapsT by augmenting the fine-grained event descriptions in AudioCaps [10] with automatically estimated onset and offset timestamps. We first use an LLM to extract individual sound events from each caption. We then use SpotSound [11] to estimate their onset and offset timestamps in the corresponding audio clip. For each audio clip, the resulting annotations are represented as

$$\mathcal{A} = \{(c_i, s_i, e_i)\}_{i=1}^N, \quad (12)$$

where N is the number of extracted events, c_i is the description of event i , and s_i and e_i are its estimated onset and offset timestamps. To the best of our knowledge, SpotSound is the best-performing model for open-vocabulary temporal grounding.

We manually evaluate 1,000 randomly sampled annotations from AudioCapsT. Of these, 841 annotations are correctly grounded, 130 contain grounding errors, and 29 reflect mismatches between the original AudioCaps captions and the audio.

4.2. Training Setup

We train DegDiT on AudioCapsT for 100 epochs. The model is optimized using AdamW with an initial learning rate of 3×10^{-4} , employing a batch size of 16 across four RTX A6000 GPUs. A linear warmup is applied for the first 400 steps, followed by linear learning-rate decay. During training, the original DegDiT architecture and objective are retained without modification.

4.3. Evaluation Setup

For evaluation, we construct a controlled event set from AudioCapsT to prevent insufficient training exposure from confounding the evaluation. We first retain events that occur at least 30 times in the dataset and further require at least 20 isolated occurrences, where the event

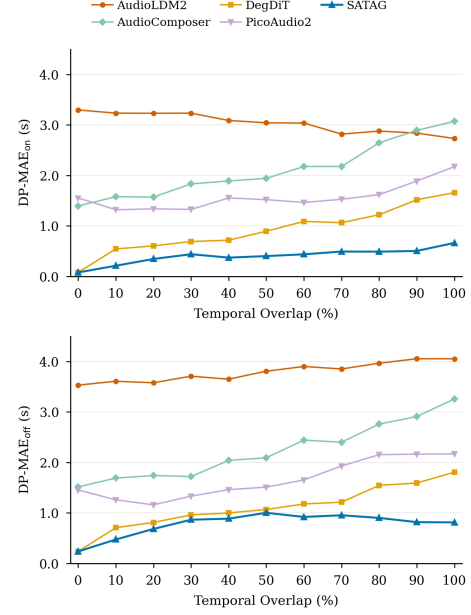


Fig. 3. Onset DP-MAE (top) and offset DP-MAE (bottom) across temporal overlap ratios for 200 event pairs.

appears without temporal overlap with other events. The filtering process results in 354 unique events, corresponding to 62,481 possible event pairs. From these pairs, we randomly sample 10,000 pairs and evaluate them under 50% and 100% overlap. The same event pairs and timestamps are used for all temporally controllable models.

We evaluate temporal generation performance using Detected Audio, Onset MAE, Offset MAE, and Detection-Penalized MAE (DP-MAE) with SpotSound. Detected Audio represents the mean number of conditioned events per sample for which SpotSound produces a valid predicted interval and is defined as

$$\text{Detected Audio} = \frac{N_{\text{detected}}}{N_{\text{samples}}}, \quad (13)$$

where N_{detected} denotes the total number of detected conditioned events and N_{samples} denotes the number of evaluated samples. Since each sample contains two conditioned events, Detected Audio ranges from 0 to 2.

For temporal evaluation, onset and offset errors are calculated only for events with valid predicted intervals. When SpotSound detects multiple intervals for the same event, intervals separated by at most 0.5 s are merged into a single interval. If multiple candidate intervals remain after merging, the interval with the largest IoU with respect to the target interval is selected, with longer duration breaking ties. Onset MAE and Offset MAE are then computed for each sample as

$$\text{Onset MAE} = \frac{1}{M} \sum_{i=1}^M |\hat{s}_i - s_i|, \quad (14)$$

$$\text{Offset MAE} = \frac{1}{M} \sum_{i=1}^M |\hat{e}_i - e_i|, \quad (15)$$

where s_i and e_i denote the target onset and offset timestamps, and $\hat{s}_i$ and $\hat{e}_i$ denote the corresponding detected timestamps. M denotes the number of events with valid predicted intervals in the sample.

Since MAE is computed only for detected events, a system can

Method	Detected Audio $\uparrow$	MAE _s $\downarrow$	MAE _e $\downarrow$	DP-MAE _{on} $\downarrow$	DP-MAE _{off} $\downarrow$
50% overlap					
AudioLDM2 [12]	0.9455	1.9339	3.1066	3.1340	3.6430
AudioComposer [3]	0.9853	0.9240	1.0398	2.7625	2.8360
PicoAudio2 [2]	1.3741	1.3223	1.2020	2.3413	2.2489
DegDiT [1]	1.6494	0.6486	1.1074	0.8746	1.2162
SATAG (ours)	1.6645	0.5430	0.9025	0.6984	0.9831
100% overlap					
AudioLDM2 [12]	0.9578	2.0650	2.7024	3.1610	3.4750
AudioComposer [3]	0.5586	0.4450	0.1966	7.3366	7.2679
PicoAudio2 [2]	1.2044	0.6284	0.6970	2.3000	2.3483
DegDiT [1]	1.4639	0.6275	0.7371	1.1044	1.1476
SATAG (ours)	1.3252	0.2546	0.3855	0.7804	0.8760

Table 1. Results at 50% and 100% overlap. Best results are in bold. DP-MAE addresses cases such as AudioComposer’s results under 100% overlap.

obtain a low error by accurately generating a small subset of the requested events. Missing events are excluded rather than counted as failures, which can favor systems with lower detection rates. To account for these failures, we also report Detection-Penalized MAE (DP-MAE) over all conditioned events. Let $\tau_i = e_i - s_i$ be the target event duration and $d_i \in \{0, 1\}$ indicate whether SpotSound detects event i with a valid interval. We define the penalized boundary errors as

$$\varepsilon_i^b = \begin{cases} \min(|\hat{t}_i^b - t_i^b|, \tau_i), & d_i = 1, \\ \tau_i, & d_i = 0. \end{cases} \quad (16)$$

Here, $b \in \{\text{on}, \text{off}\}$ denotes the boundary type, with $t_i^{\text{on}} = s_i$ and $t_i^{\text{off}} = e_i$; the predicted timestamps are defined analogously. The onset and offset DP-MAE values are averaged over all N_{eval} conditioned events in the evaluation set,

$$\text{DP-MAE}_b = \frac{1}{N_{\text{eval}}} \sum_{i=1}^{N_{\text{eval}}} \varepsilon_i^b. \quad (17)$$

DP-MAE caps boundary errors at the target event duration and assigns the same duration as the penalty for missing events or invalid intervals.

We assess perceptual audio quality and temporal accuracy using Quality MOS and Temporal Alignment MOS, respectively. Ten listeners each evaluate 50 pairs of DegDiT and SATAG clips on a 1–5 scale, yielding 500 ratings per method for each metric.

5. RESULTS

5.1. Quantitative Results

As shown in Table 1, SATAG achieves the best performance across all metrics at 50% overlap and performs well overall at 100% overlap. At 100% overlap, AudioComposer appears to perform best in terms of offset MAE alone. However, it has the highest offset DP-MAE once detection failures are penalized, indicating that its low offset MAE does not reflect reliable generation of all requested events.

Event detection is lower at 100% overlap than at 50% overlap for all methods. One possible explanation is that, at 50% overlap, the onset of the earlier event and the offset of the later event occur in regions where only one event is active, providing clear cues for temporal grounding. These isolated boundary cues are absent at 100% overlap.

Method	Quality MOS $\uparrow$	Temporal Alignment MOS $\uparrow$
DegDiT	3.250 $\pm$ 1.271	3.300 $\pm$ 1.345
SATAG (ours)	4.005 $\pm$ 1.132	4.305 $\pm$ 1.062

Table 2. Results for Quality MOS and Temporal MOS. Best results are in bold.

To further examine how existing models handle overlapping events and assess the benefit of explicit waveform superposition, we evaluate 200 event pairs at 11 overlap ratios, ranging from 0% to 100% in increments of 10 percentage points. We compute onset and offset DP-MAE at each ratio. As shown in Fig. 3, both DegDiT and SATAG achieve low errors without overlap. As overlap increases, DegDiT’s onset and offset DP-MAE generally rise, indicating a progressive decline in temporal control. In contrast, SATAG maintains relatively low errors across overlap ratios, with substantially smaller increases overall. These results support the robustness of explicit waveform superposition to increasing temporal overlap.

5.2. Subjective Results

The subjective results show consistent gains for SATAG in both perceptual quality and temporal alignment. As shown in Table 2, SATAG increases the Quality MOS from 3.250 for DegDiT to 4.005, corresponding to a relative improvement of 23.2%. Its Temporal Alignment MOS increases from 3.300 to 4.305, corresponding to a relative improvement of 30.5%. These results indicate that the improvement in temporal control does not come at the cost of perceived audio quality. Together with the objective results, these subjective evaluations support the effectiveness of waveform superposition for improving the perceived quality and temporal alignment of overlapping-event generation.

6. CONCLUSION

We propose SATAG, a superposition-aware framework for temporally controllable text-to-audio generation. Without additional training, SATAG incorporates the principle of superposition into the generation of temporally overlapping sound events. Experimental results demonstrate that our simple method achieves improved performance under overlapping-event conditions. Future work will extend the current mixing procedure by considering perceptual factors such as perceptual loudness, auditory masking, and source balance.

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